

ORAL PRESENTATION PROGRAM

Oral 1: Computer Aided Diagnosis Sept 19, 2022 (Monday), 9:00 - 10:30

Session Chairs: *Ester Bonmati, University College London, United Kingdom*
Mads Nielsen, University of Copenhagen, Denmark

- 9:00 - 9:15 Detecting Aortic Valve Pathology from the 3-Chamber Cine Cardiac MRI View**
Kavitha Vimalasvaran, Fatmatülzehra Uslu, Sameer Zaman, Christoforos Galazis, James Howard, Graham Cole, Anil A Bharath
Speaker: Kavitha Vimalasvaran, Imperial College London, United Kingdom
- 9:15 - 9:30 Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Speaker: Xinyu Liu, City University of Hong Kong, Hong Kong SAR, China
- 9:30 - 9:45 Reinforcement Learning for Active Modality Selection during Diagnosis**
Gabriel Bernardino, Anders Jonsson, Filip Loncaric, Pablo-Miki Martí Castellote, Marta Sitges, Patrick Clarysse, Nicolas Duchateau
Speaker: Gabriel Bernardino, Universite Lyon 1, France
- 9:45 - 10:00 Intra-class Contrastive Learning Improves Computer Aided Diagnosis of Breast Cancer in Mammography**
Kihyun You, Suho Lee, Kyuhee Jo, Eunhyung Park, Thijs Kooi, Hyeonseob Nam
Speaker: Kihyun You, Lunit Inc., South Korea
- 10:00 - 10:15 Contrastive Learning for Echocardiographic View Integration**
Li-Hsin Cheng, Xiaowu Sun, Rob J. van der Geest
Speaker: Li-Hsin Cheng, Leiden University Medical Center, the Netherlands
- 10:15 - 10:30 FairPrune: Achieving Fairness Through Pruning for Dermatological Disease Diagnosis**
Yawen Wu, Dewen Zeng, Xiaowei Xu, Yiyu Shi, Jingtong Hu
Speaker: Yawen Wu, University of Pittsburgh, United States

ORAL PRESENTATION PROGRAM

Oral 2: Interpretability/Uncertainty Sept 19, 2022 (Monday), 13:30 - 14:30

*Session Chairs: Aasa Feragen, Technical University of Denmark, Denmark
Pingkun Yan, Rensselaer Polytechnic Institute, United States*

- 13:30 - 13:45 **Interpretable Graph Neural Networks for Connectome-Based Brain Disorder Analysis**
Hejie Cui, Wei Dai, Yanqiao Zhu, Xiaoxiao Li, Lifang He, Carl Yang
Speaker: Hejie Cui, Emory University, USA
- 13:45 - 14:00 **What Can We Learn About a Generated Image Corrupting Its Latent Representation?**
Agnieszka Tomczak, Aarushi Gupta, Slobodan Ilic, Nassir Navab, Shadi Albarqouni
Speaker: Agnieszka Tomczak, Technical University of Munich, Germany
- 14:00 - 14:15 **Dual-graph Learning Convolutional Networks for Interpretable Alzheimer's Disease Diagnosis**
Tingsong Xiao, Lu Zeng, Xiaoshuang Shi, Xiaofeng Zhu, Guorong Wu
Speaker: Tingsong Xiao, University of Electronic Science and Technology of China, China
- 14:15 - 14:30 **Accurate and Explainable Image-based Prediction Using a Lightweight Generative Model**
Chiara Mauri, Stefano Cerri, Oula Puonti, Mark Mührlau, Koen Van Leemput
Speaker: Chiara Mauri, Technical University of Denmark, Denmark

ORAL PRESENTATION PROGRAM

Oral 3: Surgical Data Science Sept 19, 2022 (Monday), 15:30 - 17:00

Session Chairs: Sophia Bano, University College London, United Kingdom

Alberto Gomez, King's College London and Ultromics Ltd., United Kingdom

- 15:30 - 15:45 **Bayesian Dense Inverse Searching Algorithm For Real-Time Stereo Matching in Minimally Invasive Surgery**
Jingwei Song, Qiuchen Zhu, Jianyu Lin, Maani Ghaffari
Speaker: Jingwei Song, University of Michigan, United States
- 15:45 - 16:00 **Neural Rendering for Stereo 3D Reconstruction of Deformable Tissues in Robotic Surgery**
Yuehao Wang, Yonghao Long, Siu Hin Fan, Qi Dou
Speaker: Yuehao Wang, The Chinese University of Hong Kong, Hong Kong SAR, China
- 16:00 - 16:15 **Towards Holistic Surgical Scene Understanding**
Natalia Valderrama, Paola Ruiz Puentes, Isabela Hernández, Nicolás Ayobi, Mathilde Verlyck, Jessica Santander, Juan Caicedo, Nicolás Fernández, Pablo Arbeláez
Speaker: Natalia Valderrama, Universidad de los Andes, Colombia
- 16:15 - 16:30 **4D-OR: Semantic Scene Graphs for OR Domain Modeling**
Ege Özsoy, Evin Pinar Örnek, Ulrich Eck, Tobias Czempel, Federico Tombari, Nassir Navab
Speaker: Ege Özsoy, Technical University of Munich, Germany
- 16:30 - 16:45 **Video-based Surgical Skills Assessment using Long term Tool Tracking**
Mona Fathollahi, Mohammad Hasan Sarhan, Ramon Pena, Lela DiMonte, Anshu Gupta, Aishani Ataliwala, Jocelyn Barker
Speaker: Mohammad Hasan Sarhan, Johnson & Johnson Medical Ltd., Germany

ORAL PRESENTATION PROGRAM

Oral 4: Image Reconstruction & Registration Sept 20, 2022 (Tuesday), 9:00 - 10:30

Session Chairs: *Mattias Heinrich, University of Lübeck, Germany*
Tanveer Syeda-Mahmood, IBM Research, United States

- 9:00 - 9:15 **SVORT: Iterative Transformer for Slice-to-Volume Registration in Fetal Brain MRI**
Junshen Xu, Daniel Moyer, P. Ellen Grant, Polina Golland, Juan Eugenio Iglesias, Elfar Adalsteinsson
Speaker: Junshen Xu, Massachusetts Institute of Technology, United States
- 9:15 - 9:30 **NAF: Neural Attenuation Fields for Sparse-View CBCT Reconstruction**
Ruyi Zha, Yanhao Zhang, Hongdong Li
Speaker: Ruyi Zha, Australian National University, Australia
- 9:30 - 9:45 **Deep Motion Network for Freehand 3D Ultrasound Reconstruction**
Mingyuan Luo, Xin Yang, Hongzhang Wang, Liwei Du, Dong Ni
Speaker: Mingyuan Luo, Shenzhen University, China
- 9:45 - 10:00 **Towards Performant and Reliable Undersampled MR Reconstruction Via Diffusion Model Sampling**
Cheng Peng, Pengfei Guo, S. Kevin Zhou, Vishal M Patel, Rama Chellappa
Speaker: Cheng Peng, Johns Hopkins University, United States
- 10:00 - 10:15 **BMD-GAN: Bone Mineral Density Estimation Using X-Ray Image Decomposition Into Projections Of Bone-Segmented Quantitative Computed Tomography Using Hierarchical Learning**
Yi Gu, Yoshito Otake, Keisuke Uemura, Mazen Soufi, Masaki Takao, Nobuhiko Sugano, Yoshinobu Sato
Speaker: Yi Gu, Nara Institute of Science and Technology, Japan
- 10:15 - 10:30 **Fast Spherical Mapping of Cortical Surface Meshes using Deep Unsupervised Learning**
Fenqiang Zhao, Zhengwang Wu, Li Wang, Weili Lin, Gang Li
Speaker: Fenqiang Zhao, University of North Carolina at Chapel Hill, United States

ORAL PRESENTATION PROGRAM

Oral 5: Microscopy Image Analysis Sept 20, 2022 (Tuesday), 13:30 - 14:30

Session Chairs: *Yuankai Huo, Vanderbilt University, United States*
Anne Martel, Sunnybrook Research Institute, Canada

- 13:30 - 13:45 **Implicit Neural Representations for Generative Modeling of Living Cell Shapes**
David Wiesner, Julian Suk, Sven Dummer, David Svoboda, Jelmer M. Wolterink
Speaker: David Wiesner, Masaryk University, Czechia
- 13:45 - 14:00 **Gigapixel Whole-Slide Images Classification Using Locally Supervised Learning**
Jingwei Zhang, Xin Zhang, Ke Ma, Rajarsi Gupta, Joel Saltz, Maria Vakalopoulou, Dimitris Samaras
Speaker: Xin Zhang, National University of Singapore, Singapore
- 14:00 - 14:15 **Discrepancy and Gradient-guided Multi-Modal Knowledge Distillation for Pathological Glioma Grading**
Xiaohan Xing, Zhen Chen, Meilu Zhu, Yuenan Hou, Zhifan Gao, Yixuan Yuan
Speaker: Xiaohan Xing, City University of Hong Kong, Hong Kong SAR, China
- 14:15 - 14:30 **Super-Focus: Domain Adaptation for Embryo Imaging via Self-Supervised Focal Plane Regression**
Chloe He, Céline Jacques, Jérôme Chambost, Jonas Malmsten, Koen Wouters, Thomas Fréour, Nikica Zaninovic, Cristina Hickman, Francisco Vasconcelos
Speaker: Chloe He, University College London, United Kingdom

ORAL PRESENTATION PROGRAM

Oral 6: Image Segmentation Sept 20, 2022 (Tuesday), 15:30 - 17:00

Session Chairs: *Herve Lombaert, ETS Montreal, Canada*
Maria A. Zuluaga, EURECOM, France

- 15:30 - 15:45** **CACTUSS: Common Anatomical CT-US Space for US examinations**
Yordanka Velikova, Walter Simson, Mehrdad Salehi, Mohammad Farid Azampour, Philipp Paprottka, Nassir Navab
Speaker: Yordanka Velikova, Technical University of Munich, Germany
- 15:45 - 16:00** **Transformer Based Feature Fusion for Left Ventricle Segmentation In 4D Flow MRI**
Xiaowu Sun, Li-Hsin Cheng, Sven Plein, Pankaj Garg, Rob J. van der Geest
Speaker: Xiaowu Sun, Leiden University Medical Center, The Netherlands
- 16:00 - 16:15** **Self-learning and One-shot Learning based Single-slice Annotation for 3D Medical Image Segmentation**
Yixuan Wu, Bo Zheng, Jintai Chen, Danny Z. Chen, Jian Wu
Speaker: Yixuan Wu, Zhejiang University, China
- 16:15 - 16:30** **DOMINO: Domain-aware Model Calibration in Medical Image Segmentation**
Skylar E. Stolte, Kyle Volle, Aprinda Indahlastari, Alejandro Albizu, Adam J. Woods, Kevin Brink, Matthew Hale, Ruogu Fang
Speaker: Skylar E. Stolte, University of Florida, USA
- 16:30 - 16:45** **Carbon Footprint of Selecting and Training Deep Learning Models for Medical Image Analysis**
Raghavendra Selvan, Nikhil Bhagwat, Lasse F. Wolff Anthony, Benjamin Kanding, Erik B. Dam
Speaker: Raghavendra Selvan, University of Copenhagen, Denmark
- 16:45 - 17:00** **Physiology-based simulation of the retinal vasculature enables annotation-free segmentation of OCT angiographs**
Martin J. Menten, Johannes C. Paetzold, Alina Dima, Bjoern H. Menze, Benjamin Knier, Daniel Rueckert
Speaker: Martin J. Menten, Technical University of Munich and Klinikum Rechts der Isar, Germany

ORAL PRESENTATION PROGRAM

Oral 7: Surgical Planning & Simulation Sept 21, 2022 (Wednesday), 8:30 - 9:30

*Session Chairs: Sandy Engelhardt, University Hospital Heidelberg, Germany
Stamatia Giannarou, Imperial College London, United Kingdom*

- 8:30 - 8:45 **Deep Learning-based Facial Appearance Simulation Driven by Surgically Planned Craniomaxillofacial Bony Movement**
Xi Fang, Daeseung Kim, Xuanang Xu, Tianshu Kuang, Hannah H. Deng, Joshua C. Barber, Nathan Lampen, Jaime Gateno, Michael A.K. Liebschner, James J. Xia, Pingkun Yan
Speaker: Xi Fang, Rensselaer Polytechnic Institute, United States
- 8:45 - 9:00 **Ideal Midsagittal Plane Detection using Deep Hough Plane Network for Brain Surgical Planning**
Chenchen Qin, Wenxue Zhou, Jianbo Chang, Yihao Chen, Dasheng Wu, Yixun Liu, Ming Feng, Renzhi Wang, Wenming Yang, Jianhua Yao
Speaker: Wenxue Zhou, Tencent AI Lab, China
- 9:00 - 9:15 **SLAM-TKA: Real-time Intra-operative Measurement of Tibial Resection Plane in Conventional Total Knee Arthroplasty**
Shuai Zhang, Liang Zhao, Shoudong Huang, Hua Wang, Qi Luo, Qi Hao
Speaker: Shuai Zhang, University of Technology Sydney, Australia
- 9:15 - 9:30 **Deep Geometric Supervision Improves Spatial Generalization in Orthopedic Surgery Planning**
Florian Kordon, Andreas Maier, Benedict Swartman, Maxim Privalov, Jan S. El Barbari, Holger Kunze
Speaker: Florian Kordon, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

ORAL PRESENTATION PROGRAM

Oral 8: Integration Beyond Imaging Sept 21, 2022 (Wednesday), 13:30 - 15:00

Session Chairs: *Kayhan Batmanghelich, University of Pittsburgh, United States*
Ruogu Fang, University of Florida, United States

- 13:30 - 13:45** **Tagged-MRI Sequence to Audio Synthesis via Self Residual Attention Guided Heterogeneous Translator**
Xiaofeng Liu, Fangxu Xing, Jerry L. Prince, Jiachen Zhuo, Maureen Stone, Georges El Fakhri, Jonghye Woo
Speaker: Xiaofeng Liu, MGH and Harvard, United States
- 13:45 - 14:00** **TranSQ: Transformer-based Semantic Query for Medical Report Generation**
Ming Kong, Zhengxing Huang, Kun Kuang, Qiang Zhu, Fei Wu
Speaker: Ming Kong, Zhejiang University, China
- 14:00 - 14:15** **Multimodal-GuideNet: Gaze-Probe Bidirectional Guidance in Obstetric Ultrasound Scanning**
Qianhui Men, Clare Teng, Lior Drukker, Aris T. Papageorghiou, J. Alison Noble
Speaker: Qianhui Men, University of Oxford, United Kingdom
- 14:15 - 14:30** **Fusing Modalities by Multiplexed Graph Neural Networks for Outcome Prediction in Tuberculosis**
Niharika S. D'Souza, Hongzhi Wang, Andrea Giovannini, Antonio Foncubierta-Rodriguez, Kristen Beck, Orest Boyko, Tanveer Syeda-Mahmood
Speaker: Tanveer Syeda-Mahmood, IBM Research, United States
- 14:30 - 14:45** **Identify Consistent Imaging Genomic Biomarkers for Characterizing the Survival-associated Interactions between Tumor-infiltrating Lymphocytes and Tumors**
Yingli Zuo, Yawen Wu, Zixiao Lu, Qi Zhu, Kun Huang, Daoqiang Zhang, Wei Shao
Speaker: Yingli Zuo, Nanjing University of Aeronautics and Astronautics, China
- 14:45 - 15:00** **Extended Electrophysiological Source Imaging with Spatial Graph Filters**
Feng Liu, Guihong Wan, Yevgeniy R. Semenov, Patrick L. Purdon
Speaker: Feng Liu, Stevens Institute of Technology, United States